

Capstone Project Proposal Report

Individual Report

VIT-AP University, School of Computer Science and Engineering

Guide Approval (initials/date): Dr Battula Venkata Satish Babu

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STUDENT INFORMATION

Student Name	Lingala Sandeep Goud
Student Register Number	22MIS7204
Programme	Integrated M.Tech - Software Engineering
Semester / Year	Fast Track 2026-27 / 5th Year
Guide(s)	Dr. Battula Venkata Satish Babu, VIT-AP University
Project Title	Implementation of a Hybrid Post-Quantum Cryptographic Communication System Using ML-KEM and AES-256

TEAM COMPOSITION

Provide the information below for each member of the project team. Include all project team members, not just those in your discipline or those enrolled for Capstone project. Please also include yourself!

Reg. No	Name	Major	Specialization
22MIS7235	Shaik Ibrahim Khalilulla	Integrated M.Tech SE	Cyber Security & Post-Quantum Cryptography
22MIS7239	Bommanaboina Yaswanth	Integrated M.Tech SE	Cyber Security & Cryptography
22MIS7204	Lingala Sandeep Goud	Integrated M.Tech SE	Backend Development & Post-Quantum Cryptography
22MIS7016	S B Mohammad Afzal	Integrated M.Tech SE	Frontend Development

PROJECT AND TASK DESCRIPTION

1. What the Project is:

Traditional secure communication systems rely on public-key cryptographic algorithms such as RSA and Elliptic Curve Cryptography (ECC), which are expected to become vulnerable once large-scale quantum computers become available. Quantum algorithms like Shor's Algorithm can efficiently break these classical encryption techniques, posing a significant threat to data confidentiality and secure communications.

This project aims to develop a Hybrid Post-Quantum Cryptographic Communication System that combines ML-KEM (Module Lattice-Based Key Encapsulation Mechanism) with AES-256 symmetric encryption to provide secure file sharing and real-time messaging.

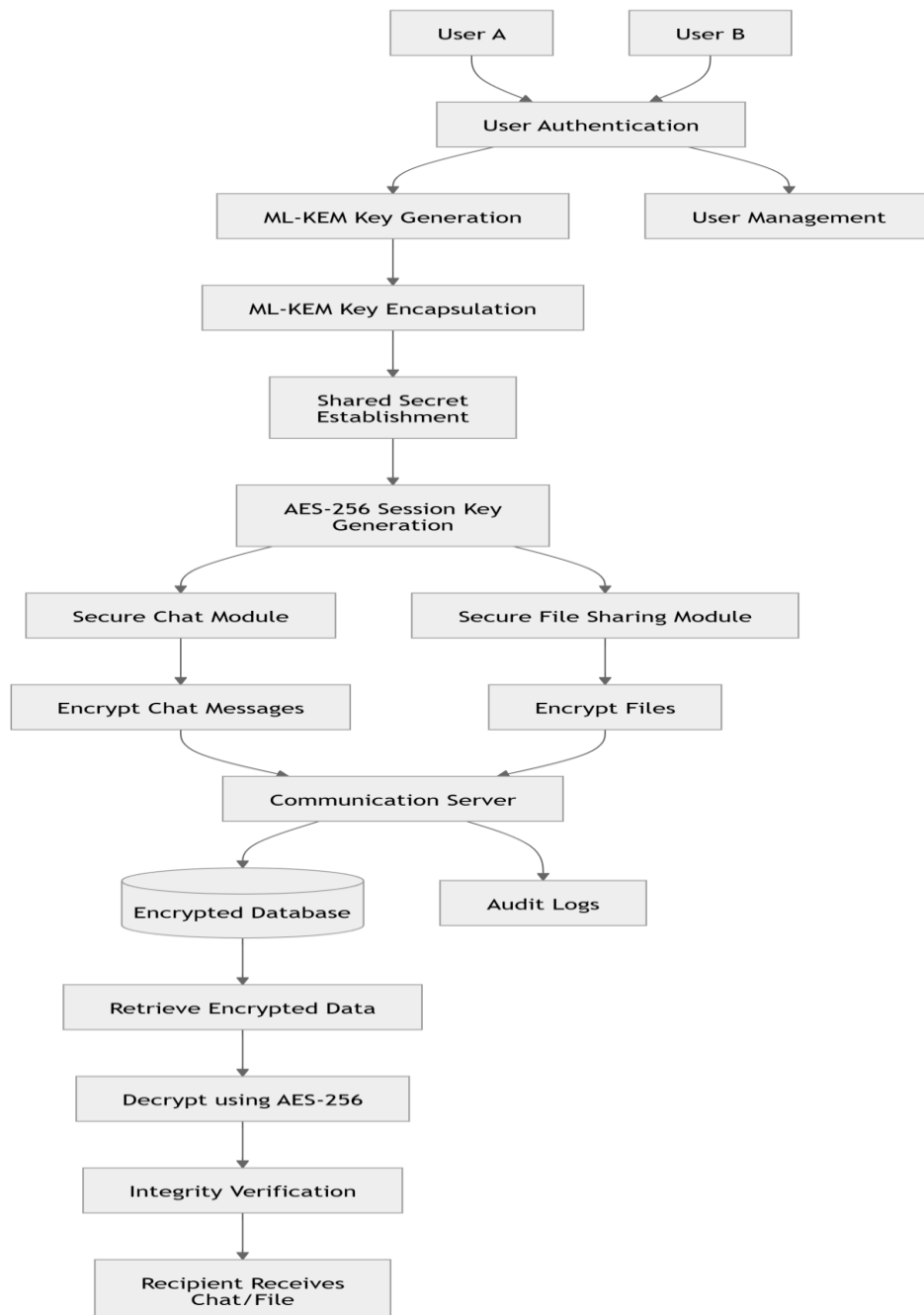
The proposed system establishes a secure communication channel using ML-KEM to exchange encryption keys securely. Once the shared secret is established, AES-256 is used for fast encryption and decryption of files and chat messages. The platform will provide end-to-end encryption, ensuring confidentiality, integrity, and resistance against both classical and future quantum attacks. The project focuses on building a practical communication platform capable of securely exchanging files and instant messages while maintaining high performance, scalability, and usability.

2. How it Works:

The proposed Hybrid Post-Quantum Cryptographic Communication System provides secure real-time messaging and file sharing by combining ML-KEM (Module Lattice-Based Key Encapsulation Mechanism) with AES-256 encryption. The communication process begins with user registration and authentication, where users securely log into the system. After successful authentication, the ML-KEM Key Generation and Key Encapsulation modules establish a secure shared secret between the sender and receiver without exposing any sensitive key material. This shared secret is then used to generate an AES-256 session key, which enables fast and secure encryption of chat messages and files.

The Communication Server acts as an intermediary that transfers encrypted messages and files while storing only encrypted data in the Encrypted Database, preventing unauthorized access to sensitive information. The system also maintains Audit Logs to record important activities such as user authentication, file transfers, and communication events. By integrating ML-KEM for quantum-resistant key exchange and AES-256 for efficient symmetric encryption, the proposed architecture provides a secure, scalable, and future-proof communication platform.

System Architecture:



3. Individual Roles and Specific Tasks:

Member	Responsibilities
Shaik Ibrahim Khalilulla (22MIS7235)	Project Lead, System Architecture Design, ML-KEM Integration, Overall Backend Development, Documentation, Research Paper Preparation
Lingala Sandeep Goud (22MIS7204)	Real-Time Chat Module using WebSockets, Authentication Module, API Development, Database Integration
Bommanaboina Yaswanth (22MIS7239)	AES-256 Encryption Module, Secure File Sharing Module, Key Management, Performance Testing
S B Mohammad Afzal (22MIS7016)	Frontend Development, UI/UX Design, Testing, Deployment, Report Preparation and Presentation

4. Technical Approach:

The system employs a hybrid cryptographic architecture. ML-KEM (Kyber) handles quantum-resistant key encapsulation during the handshake phase, generating a shared secret without transmitting any private key material. This shared secret is then used to derive an AES-256 session key for symmetric encryption of all subsequent communication.

The backend is built using Python (FastAPI/Flask) with WebSocket support for real-time communication. The frontend is developed in React.js. Authentication is handled via JWT tokens. The encrypted data is stored in MySQL/SQLite databases, and all activity is logged through an audit trail system. The entire stack is containerized using Docker and deployed via Nginx.

Suggested Technologies:

Component	Technology
Frontend	React.js
Backend	Python FastAPI / Flask
Database	MySQL / SQLite
Real-time Chat	WebSockets / Socket.IO
Encryption	AES-256
Post-Quantum Cryptography	ML-KEM (Kyber)
Authentication	JWT
Deployment	Docker + Nginx
Version Control	Git & GitHub

5. Time Plan:

Stage	Timeline	Deliverables
First Review	June	Problem Statement, Literature Survey, System Design, Initial ML-KEM Implementation
Second Review	July	Secure Chat Module, File Encryption Module, AES Integration, Database Development
Final Review	August	Complete Working System, Performance Evaluation, Testing Results
Final Submission	31st August	Project Report, Presentation, Demonstration, Research Paper Draft

OUTCOME MATRIX

Outcome	Plan for Demonstrating Outcome
a) Apply mathematics, science and engineering	Implement lattice-based cryptographic algorithms, secure key exchange, AES encryption, and evaluate cryptographic performance.
b) Design within realistic constraints	Develop a scalable secure communication system considering computation cost, usability, storage, and security.
c) Function on multidisciplinary teams	Four-member team working on backend, frontend, security, cryptography, and deployment.
d) Identify and solve engineering problems	Design a practical quantum-resistant communication system addressing secure messaging and secure file transfer challenges.
e) Communicate effectively	Project reviews, technical documentation, presentations, demonstration, and final report.
f) Use modern engineering tools	Python, FastAPI/Flask, React.js, ML-KEM Library, AES-256, SQLite/MySQL, WebSockets, GitHub.

REALISTIC CONSTRAINTS

- Computational overhead introduced by lattice-based cryptographic algorithms.
- Limited computing resources during encryption and key generation.
- Network latency affecting real-time communication.
- Secure storage and management of cryptographic keys.
- Maintaining compatibility with existing communication infrastructure.
- Scalability for multiple concurrent users.
- Project completion within the allocated semester duration.

ENGINEERING STANDARDS

- NIST Post-Quantum Cryptography Standard (ML-KEM)
- FIPS 197 AES-256 Encryption Standard
- OWASP Secure Coding Guidelines
- TLS Security Concepts
- REST API Design Standards
- Git Version Control Best Practices
- IEEE Software Engineering Standards
- Secure Software Development Life Cycle (SSDLC)